PEER REVIEW HISTORY

BMJ Open publishes all reviews undertaken for accepted manuscripts. Reviewers are asked to complete a checklist review form (**http://bmjopen.bmj.com/site/about/resources/checklist.pdf**) and are provided with free text boxes to elaborate on their assessment. These free text comments are reproduced below.

ARTICLE DETAILS

TITLE (PROVISIONAL)	Risk of aspiration pneumonia in pediatric patients with dysphagia who were found to have laryngeal penetration on the instrumental swallow evaluation. A systematic review protocol
AUTHORS	Adlakha, Vaishali; Ramos, Leona; Smith, Abigail; Tsistinas, Olivia; Tanner-Smith, Emily; Merrow, Jill; Soultan, Zafer; Mortelliti, Anthony; Rosen, Rachel; Imdad, Aamer

VERSION 1 – REVIEW

REVIEWER	Raol, Nikhila Emory University School of Medicine
REVIEW RETURNED	12-Mar-2021

GENERAL COMMENTS	The authors present a systematic review/meta-analysis protocol to look at the risk of aspiration in children with penetration of videofluoroscopic swallow study. I think this study is necessary and will be a good addition to the literature. I have some concerns: -Excluding patients with anatomic reasons: Can you explain further what you mean by this? Not all patients with anatomic abnormalities that could cause dysphagia (i.e. cleft) actually present with dysphagia. In addition, diagnosis of an anatomic abnormality often requires exam under anesthesia so it may be tough to snuff out those patients. I would recommend removing this as an exclusion criteria. -Mild, moderate, and severe penetration is not a standard way of describing degree of dysphagia. Consider using the penetration aspiration scale or deep vs. shallow, wide column vs. narrow column, etc. -Consider making pneumonia a binary outcome as it will likely be difficult to distinguish severe PNA in the majority of articles. -Consider including functional endoscopic evaluation of the swallow as another modality for evaluating penetration. -What about other special populations for subgroup analysis? i.e. Congenital cardiac disease, prematurity, syndromic children I look forward to your revisions and to the outcomes of this systematic review/meta-analysis.

REVIEWER	Hirsch, Alexander Harvard Medical School
REVIEW RETURNED	15-Apr-2021

GENERAL COMMENTS	This is an important and worthwhile endeavor and I believe this work will contribute to our understanding of pediatric aspiration pneumonia. I worry a little about the heterogeneous definitions of pneumonia you may encounter in the literature review but your approach seems like a reasonable way to tackle this.

VERSION 1 – AUTHOR RESPONSE

Reviewer #1:

The authors present a systematic review/meta-analysis protocol to look at the risk of aspiration in children with penetration of video fluoroscopic swallow study. I think this study is necessary and will be a good addition to the literature.

Response: Thank you for your kind feedback.

Comments/questions for the authors:

1. Excluding patients with anatomic reasons: Can you explain further what you mean by this? Not all patients with anatomic abnormalities that could cause dysphagia (i.e. cleft) actually present with dysphagia. In addition, diagnosis of an anatomic abnormality often requires exam under anesthesia so it may be tough to snuff out those patients. I would recommend removing this as an exclusion criteria.

Response: Thank you for the thoughtful comment. We agree with the reviewer and have expanded the text in this section as follows:

"We will consider studies that include a mixed population with and without anatomical problems. We will also include studies where the anatomical problem was not known prior to performing VFSS. We will however exclude studies that focused exclusively on patients with a particular anatomical anomaly of the aerodigestive tract (e.g. cleft palate etc) or a particular medical diagnosis (e.g. Mysthenia gravis, post-stroke etc) as the results from those studies may not be generalizable to the population of interest.".

2. Mild, moderate, and severe penetration is not a standard way of describing degree of dysphagia. Consider using the penetration aspiration scale or deep vs. shallow, wide column vs. narrow column, etc.

Response: Thank you. We agree with this comment. We have modified the text to include the aspiration penetration scale to describe degree of penetration. We also included additional

classification of laryngeal penetration and cite the relevant reference for these classifications. The text has been modified as follows:

"We will consider exposure as yes or no (any laryngeal penetration vs. no laryngeal penetration for any consistency) and also consider the degrees of penetration. The degree of laryngeal penetration will be categorized using the 8 point aspiration penetration scale as follows:

Material does not enter airway.

Material enters the airway, remains above the vocal folds, and is ejected from the airway.

Material enters the airway, remains above the vocal folds, and is not ejected from the airway.

Material enters the airway, contacts the vocal folds, and is ejected from the airway.

Material enters the airway, contacts the vocal folds, and is not ejected from the airway.

Material enters the ariway, passes below the vocal folds, and is ejected into the larynx or out of the airway.

Material enters the airway, passes below the vocal folds, and is not ejected from the trachea despite effort.

Material enters the airway, passes below the vocal folds, and no effort is made to eject.

We will also consider categorizing exposure based on mild/None vs. moderate vs. deep laryngeal penetration. The definition of none/mild vs. Moderate vs. deep laryngeal penetration will be based on definitions given in reference 8:

-No/Mild laryngeal penetration: penetration of contrast medium under the tip of the epiglottis;

-Moderate penetration: entry of contrast medium into the upper two-thirds of the laryngeal vestibule and remaining above the arbitrary plane that extends from the top of the arytenoids to the top of the thyroid cartilage and;

Deep laryngeal penetration: entry of contrast medium into the lower one-third of the laryngeal vestibule below the arbitrary plane that extends from the top of the arytenoids to the top of the thyroid cartilage; this level of laryngeal penetration may or may not result in coating of the vocal cords.

We will consider the penetration score discussed above: Score 0-3 as mild/none, score 4: Moderate and score 5: deep laryngeal penetration.

3. Consider making pneumonia a binary outcome as it will likely be difficult to distinguish severe PNA in the majority of articles.

Response: Thank you for your comment. We intended to measure the pneumonia outcome as dichotomous outcome. We further clarify this in the manuscript as follows. “Incidence of aspiration pneumonia: Dichotomous outcome measured at 3 months, 6 months, and at the longest follow-up”. We further mention the type of outcome (Dichotomous or continuous) for each of the secondary outcomes as well

4. Consider including functional endoscopic evaluation of the swallow as another modality for evaluating penetration.

Response: We agree with this comment and have included the same in our title and methodology as follows:

“TITLE: Risk of aspiration pneumonia in pediatric patients with dysphagia who were found to have laryngeal penetration on the instrumental swallow evaluation. A systemic review protocol”

“Population will include pediatric patients (<18 years) with dysphagia who have been evaluated with VFSS or Fibreoptic endoscopic evaluation of swallowing (FEES).”

5. What about other special populations for subgroup analysis? i.e. Congenital cardiac disease, prematurity, syndromic children.

Response: Thank you. We agree with reviewer’s comment. We include the following additional subgroup analysis as noted now:

Subgroup analysis and investigation of heterogeneity

1) Study population: Age: Infants (< 1 year of age) vs. children (1-5 years of age)
2) Study population: Neurological comorbidity: Children with CNS anomalies vs. children with no known CNS anomalies
3) Study population: Syndromic comorbidity: Children with a syndromic diagnosis vs. children with no syndromic diagnosis
4) Study population: Anatomic anomalies of airway or gastrointestinal tract: Children with anatomical anomalies vs. children with no anatomical anomalies.
5) Exposure: Aspiration scale score: Scores of 0-2 vs. 3-5 vs. 6-8
6) Exposure: Laryngeal penetration: None/mild vs. Moderate vs. Deep penetration
7) Exposure: Consistency of food: Thin liquid vs. thick liquid vs. Puree vs. solid food

These subgroup analyses will be considered based on subgroup effects reported in studies based on individual patient features

Reviewer #2

1. This is an important and worthwhile endeavor and I believe this work will contribute to our understanding of pediatric aspiration pneumonia. I worry a little about the heterogeneous definitions of pneumonia you may encounter in the literature review but your approach seems like a reasonable way to tackle this.

Response: Thank you. We agree with this comment. We anticipate this problem and have elaborated it in the text as follows *"The authors of the included studies may not follow a uniform definition of pneumonia. We will assess the definition of pneumonia in each of the included studies and see if they followed the WHO definition. If a study does not follow the WHO definition, we will note that but still include the data from that study as reported by the authors. Further if the authors report the data on pneumonia based on severity (like mild, moderate, and severe), and these categories are mutually exclusive, in that case, we will combine these categories to make one outcome (Pneumonia, Yes or No)."*

Further, we have included an additional category in sensitivity analysis *"WHO definition of pneumonia vs. nonstandard definition of pneumonia".*

VERSION 2 – REVIEW

REVIEWER	Raol, Nikhila Emory University School of Medicine
REVIEW RETURNED	05-Jul-2021

GENERAL COMMENTS	Thank you for these edits. I believe it is ready for publication and I look forward to the accompanying resulting manuscript.